

Practice 2

“On my honor, as an Aggie, I have neither given nor received unauthorized aid on this academic work.”

Signature:

Name:

You may use the following list of mathematical expressions.

Definition. Let V be a vector space over F . A *subspace* of V is a subset $W \subset V$ which is itself a vector space over F .

Theorem. A non-empty subset $W \subset V$ is a subspace of V if and only if for every pair of vector $\mathbf{w}_1, \mathbf{w}_2 \in W$ and every scalar $s \in F$ the vector $s\mathbf{w}_1 + \mathbf{w}_2$ is again in W .

Definition. A *norm* on vector space V is a function $\|\cdot\| : V \rightarrow \mathbb{R}$ that satisfies

1. $\|\mathbf{v}\| \geq 0 \quad \forall \mathbf{v} \in V$; equality holds if and only if $\mathbf{v} = \mathbf{0}$ (positive definiteness)
2. $\|s\mathbf{v}\| = |s| \|\mathbf{v}\| \quad \forall \mathbf{v} \in V, s \in F$ (positive homogeneity)
3. $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\| \quad \forall \mathbf{v}, \mathbf{w} \in V$. (triangle inequality)

Definition. Let F be the field of real numbers or the field of complex numbers, and let V be a vector space over F . An *inner product* on V is a function which assigns to each ordered pair of vectors $\mathbf{v}, \mathbf{w} \in V$ a scalar $\langle \mathbf{v} | \mathbf{w} \rangle \in F$ in such a way that for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all scalars $s \in F$

1. $\langle \mathbf{u} + \mathbf{v} | \mathbf{w} \rangle = \langle \mathbf{u} | \mathbf{w} \rangle + \langle \mathbf{v} | \mathbf{w} \rangle$
2. $\langle s\mathbf{v} | \mathbf{w} \rangle = s \langle \mathbf{v} | \mathbf{w} \rangle$
3. $\langle \mathbf{v} | \mathbf{w} \rangle = \overline{\langle \mathbf{w} | \mathbf{v} \rangle}$, where the overbar denotes complex conjugation;
4. $\langle \mathbf{v} | \mathbf{v} \rangle > 0$ if $\mathbf{v} \neq \mathbf{0}$.

Definition. Consider a complex $n \times n$ matrix A with elements a_{ij} . The Hermitian transpose $B = A^H$ of A is defined by the elements $b_{ij} = \overline{a_{ji}}$ where $\bar{a}$ denotes the complex conjugate of a .

Definition. A complex matrix A is *Hermitian* if $A^H = A$. It is *skew Hermitian* if $A^H = -A$.

Theorem. Let V and W be vector spaces over the field F and let T be a linear transformation from V into W . If V is finite-dimensional, then $\text{rank}(T) + \text{nullity}(T) = \text{dimension}(V)$.

Theorem. Let V be a vector space and $T : V \rightarrow V$ be a projection operator. Then, the range $\mathcal{R}(T)$ and the $\mathcal{N}(T)$ are disjoint subspaces of V .

Problems:

1. Let V be a vector space over the real numbers. Consider the ordered basis $\mathcal{A} = (\underline{v}_1, \underline{v}_2, \underline{v}_3)$, and linear transformation $T : V \rightarrow V$ such that

$$T\underline{v}_1 = 2\underline{v}_1 + \underline{v}_2 - \underline{v}_3$$

$$T\underline{v}_2 = -2\underline{v}_1 - \underline{v}_2 + 2\underline{v}_3$$

$$T\underline{v}_3 = \underline{v}_3$$

- (a) **1 pt** – Find matrix representation A for this linear transformation under ordered basis $\mathcal{A}$; that is, find matrix A such that $[T\underline{v}]_{\mathcal{A}} = A [\underline{v}]_{\mathcal{A}}$.

- Matrix $A =$

- (b) **1 pt** – Find the rank and the nullity of linear transformation T .

- Rank:
- Nullity:

- (c) **1 pt** – Consider ordered basis $\mathcal{B} = (\underline{w}_1, \underline{w}_2, \underline{w}_3)$, where

$$\underline{w}_1 = 2\underline{v}_1 + \underline{v}_2$$

$$\underline{w}_2 = \underline{v}_1 + \underline{v}_2$$

$$\underline{w}_3 = \underline{v}_3$$

Find matrix M such that $[\underline{v}]_{\mathcal{A}} = M [\underline{v}]_{\mathcal{B}}$ and show that $\mathcal{B}$ is a basis.

- Matrix $M =$

- (d) **1 pt** – Find matrix representation B for linear transformation T under ordered basis $\mathcal{B}$; that is, find matrix B such that $[T\underline{v}]_{\mathcal{B}} = B [\underline{v}]_{\mathcal{B}}$.

- Matrix $B =$

- (e) **1 pt** – Show that B is a projection matrix.

2. Let V be the vector space $\mathbb{R}^3$ with the inner product defined by

$$\langle \mathbf{v}_1 | \mathbf{v}_2 \rangle_M = \mathbf{v}_2^T M \mathbf{v}_1 = \mathbf{v}_2^T \begin{bmatrix} 1 & 0 & 1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1 \end{bmatrix} \mathbf{v}_1.$$

Similarly, let W be the vector space $\mathbb{R}^2$ with the standard inner product $\langle \mathbf{w}_1 | \mathbf{w}_2 \rangle = \mathbf{w}_2^T \mathbf{w}_1$. Consider the linear transformation $A \in L(V, W)$ given by

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Take a close look at the inner product on V before you start answering questions, it is not the standard inner product.

- (a) **0.5 pt** – Identify the range R_A and the null space N_A of the linear transformation A .
- (b) **1 pt** – Starting from the standard basis, use the Gram-Schmidt process to find an orthogonal basis for V .
- (c) **1 pt** – Find the matrix P_{N_A} that projects V on the null space of A . Also, compute the projection matrix $P_{N_A^\perp}$, where $N_A^\perp$ is the orthogonal complement of N_A . Hint: We are not using the standard inner product; you cannot use the standard formula for the projection matrix.
- (d) **0.5 pt** – Show that $\langle \mathbf{v}_1 | P_{N_A} \mathbf{v}_2 \rangle_M = \langle P_{N_A} \mathbf{v}_1 | \mathbf{v}_2 \rangle_M$.
- (e) **1 pt** – The adjoint of A is a matrix A^* such that $\langle A\mathbf{v} | \mathbf{w} \rangle = \langle \mathbf{v} | A^* \mathbf{w} \rangle_M$ for all $\mathbf{v} \in V$ and $\mathbf{w} \in W$. Find the adjoint of A .
- (f) **1 pt** – Compute the operator norm of A .